$$x = x_0 + v_0 + \frac{1}{2}at^2, \theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$$
$$Galilean: t' = t, x' = x - vt$$
$$Lorentz: ct' = \gamma ct - \gamma\beta x$$
$$x' = \gamma x - \gamma\beta ct$$
$$Inv\ Lorentz: ct = \gamma ct' + \gamma\beta x'$$
$$x = \gamma x' + \gamma\beta ct'$$

$$(\Delta s)^2 = (\Delta ct)^2 - (\Delta x)^2 = (\Delta ct')^2 - (\Delta x')^2 = (\Delta s')^2$$
$$c = 2.998 \times 10^8 ms^{-1}$$
$$m_{Relativistic} = \gamma m$$
$$v = \frac{v' + \beta c}{c + \beta v'}c,$$
$$Vel\ v' in\ rest\ frame\ and\ v\ in\ stationary\ frame$$
$$KE = mc^2(\gamma - 1),\ E_{Total} = KE + mc^2 = \gamma mc^2$$
$$v = \lambda f = \lambda v = c$$
$$\theta_i = \theta_r,\ \theta_c = arcSin(\frac{n_1}{n_2})$$
$$f = \frac{n_2 r}{n_2 - n_1},\ f_{Sphr\ Mirr} = \frac{r}{2}$$
$$\frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, m' = \frac{h_i}{h_o} = \frac{d_i}{d_o},$$
$$2t = \left(m + \frac{s}{2}\right)\lambda, Constructive\ reflection\ (s\ phase\ shifts)$$
$$Sin(\theta) = \frac{m\lambda}{a}, Single\ Slit\ (minima)$$
$$Sin(\theta) = \frac{m\lambda}{d}, Aperture\ (minima)$$
$$I = I_{Max}\big(Cos(\beta)\big)^2\left(\frac{Sin(\alpha)}{\alpha}\right)^2, Multislit, b\ is\ slit\ spacing$$
$$m\lambda = dSin(\theta), Maxima$$

$$\gamma = \frac{\delta}{2}, \alpha = \frac{\pi a}{\lambda}Sin(\theta), \delta = \frac{2\pi}{\lambda}\Delta r = \frac{2\pi d}{\lambda}Sin(\theta)$$
$$x = \frac{lm\lambda}{\sqrt{b^2 - m^2\lambda^2}}, m^{th} Bright\ Two\ slit$$

$$I_s(\lambda, T)d\lambda = \frac{2\pi hc^2}{\lambda^5}\frac{1}{e^{\frac{hc}{\lambda kT}} - 1}d\lambda \quad [Js^{-1}m^{-2}m^{-1}]$$
$$\lambda_{max}T = 2.898 \times 10^{-3}\ [mK], \qquad E = hv$$

$$p = \frac{E}{c} = \frac{hv}{c} = \frac{h}{\lambda}, \lambda = \frac{h}{p} = \frac{h}{\gamma m_0 v}$$
$$\frac{hc}{\lambda} = \frac{hc}{\lambda'} + K = \frac{hc}{\lambda'} + (\gamma - 1)m_0c^2$$

$$\frac{h}{\lambda} = \frac{h}{\lambda'}Cos(\phi) + \gamma m_0 v Cos(\theta)\ \{x\}$$
$$0 = \frac{h}{\lambda'}Sin(\phi) - \gamma m_0 v Sin(\theta)\ \{y\}$$
$$k = \frac{2\pi}{\lambda} = \frac{p}{\hbar} = \frac{\sqrt{2mE}}{\hbar}(Angular\ Wavenumber)$$
$$T = \frac{16k^2b^2}{(b^2 + k^2)^2}e^{-2bl} \approx \left[\frac{16E}{U}\left(1 - \frac{E}{U_b}\right)\right]e^{-2bl}$$

$$\infty\ U\ Well: kl = n\pi, k = \frac{\sqrt{2mE}}{\hbar}$$

$$v = v_0 + at, \omega = \omega_0 + \alpha t$$

$$v^2 = v_0^2 + 2a(x - x_0), \qquad \omega^2 = \omega^2 + 2\alpha(\theta - \theta_0)$$
$$\beta = \frac{v}{c}, \gamma = \frac{1}{\sqrt{1 - \beta^2}}$$
$$\Delta t = \gamma\Delta t_{Proper}, t_{Proper} is\ shortest$$
$$\Delta x = \frac{1}{\gamma}\Delta x_{Proper}, x_{proper}\ is\ longest$$
$$\Delta s^2 < 0 - Spacelike$$
$$\Delta s^2 = 0 - light\ like$$
$$\Delta s^2 > 0 - Time\ like$$
$$v' = \frac{v - \beta c}{c - \beta v}c,$$
$$Vel\ v\ in\ rest\ frame\ and\ v'\ in\ moving\ frame$$
$$p = \gamma mv = \sqrt{\gamma^2 - 1}mc, \therefore E^2 = (pc)^2 + (mc^2)^2$$
$$E_{Total} = E_{Absorbed} + E_{Reflected} + E_{Transmitted}$$
$$n_1 Sin(\theta_1) = n_2 Sin(\theta_2)$$

$$\frac{n_1}{d_0} + \frac{n_2}{d_i} = \frac{n_2 - n_1}{r}, \qquad \frac{1}{f} = (n_{12} - 1)\left(\frac{1}{r_1} - \frac{1}{r_2}\right), (n_{12} = \frac{n_2}{n_1})$$
$$\phi = \pi: n_1 < n_2, \phi = 0: n_1 > n_2$$

$$2t = \left(m + \frac{s + 1}{2}\right)\lambda, Destructive\ reflection$$
$$I = I_{Max}\left(\frac{Sin(\alpha)}{\alpha}\right)^2, \alpha = \frac{\phi}{2} = \frac{\pi a}{\lambda}Sin(\theta)$$
$$\theta_c = 1.22\frac{\lambda}{d}$$
$$\beta = \frac{\pi b}{\lambda}Sin(\theta), \alpha = \frac{\pi a}{\lambda}Sin(\theta)$$
$$I = I_0\left(\frac{Sin(\alpha)}{\alpha}\right)^2\left(\frac{Sin(N\gamma)}{Sin(\gamma)}\right)^2$$
$$\frac{\lambda}{\Delta\lambda} = mN = \frac{mw}{d}$$
$$x = \frac{l(m - \frac{1}{2})\lambda}{\sqrt{b^2 - (m - \frac{1}{2})^2\lambda^2}}, m^{th} Dark\ Two\ slit$$

$$I = \sigma T^4,\ 5.67 \times 10^{-8},\ \hbar = \frac{h}{2\pi},\ L = \sigma T^4 4\pi R^2$$

$$\lambda = \frac{h}{p} = \frac{h}{\gamma m_0 v}, 1.602 \times 10^{-19}\left[\frac{J}{eV}\right], K_{max} = hv - \phi,$$
$$\Delta = \lambda' - \lambda = \frac{h}{m_0 c}(1 - Cos(\phi))$$
$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$
φ = photon, θ = electron

$$|\psi|^2 = \left|\psi_0 e^{-ikx}\right|^2 = \psi_0^2(e^{-ikx})(e^{-ikx})^* = \psi_0^2 = Const$$

$$\Delta x\Delta p \geq \frac{\hbar}{2}\ or\ \Delta E\Delta t \geq \frac{\hbar}{2}$$

$$Where\ b = \sqrt{\frac{2m(U_b - E)}{\hbar^2}}$$

$$\psi_n(x) = \sqrt{\frac{2}{l}}Sin\left(\frac{n\pi}{l}x\right)$$

$$E_n = \left(\frac{\hbar^2\pi^2}{2ml^2}\right)n^2 = \left(\frac{h^2}{8ml^2}\right)n^2$$
$$L = n\hbar$$

$$E_n = -\left(\frac{m_e e^4}{8\epsilon_0^2 h^2}\right)\frac{1}{n^2} = -\frac{13.6eV}{n^2}$$
$$E_{n_x,n_y,n_z} = \frac{h^2}{8m}\left(\frac{n_x^2}{l_x^2} + \frac{n_y^2}{l_y^2} + \frac{n_z^2}{l_z^2}\right)$$
$$n - Principle\ QN\ (E_n), \ell - Orbital\ QN$$

$$m_\ell \in (-\ell, \ell) - Orbital\ Magnetic\ QN$$

$$s - Spin\ \ QN\ \left(\frac{1}{2} for\ pne\right), m_s = \frac{1}{2}, -\frac{1}{2} - spin\ QN$$
$$L = \sqrt{\ell(\ell+1)}\hbar$$
$$\mu_{orb} = -\frac{\ell}{2m}L = -\frac{\ell}{2m}\sqrt{\ell(\ell+1)}\hbar$$
$$\mu_{orb,z} = -\frac{\ell}{2m}L_z = -\frac{\ell}{2m}m_\ell\hbar = -m_\ell\mu_B$$
$$\mu_B = \frac{\ell\hbar}{2m} = 9.724 \times 10^{-24} JT^{-1}$$
$$For\ a\ given\ L\ there\ will\ be\ 2\ell + 1\ L_z$$

$$Bremstrahlung: \frac{1}{\lambda_{K\alpha}} = \frac{3}{4}R(z-1)^2\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right),$$
$$z = Atomic\ Num$$

$$\beta^-: {}^A_Z X \rightarrow {}^A_{z+1}Y + e^- + \bar{\nu} + E$$
$$\beta^+: {}^A_Z X \rightarrow {}^A_{z-1}Y + e^+ + \nu + E$$
$$R \equiv -\frac{dN}{dt} = \lambda N = R_0 e^{-\lambda t}$$
$$\vec{F} = \frac{kq_1q_2}{r^2}\hat{r} = \vec{E}q, |\vec{F}| = \frac{1}{4\pi\epsilon_0}\frac{|q_1||q_2|}{r^2}$$
$$\epsilon_0 = 8.55 \times 10^{-12} C^2 N^{-1} m^{-2}$$

$$\vec{p} = q\vec{d},\ \ E_{Dipole} = \frac{1}{2\pi\epsilon_0}\frac{p}{z^3},\ \ \vec{E} = \frac{\sigma}{2\epsilon_0}$$
$$V = \frac{U}{q} = -\int_1^2 \vec{E}\cdot ds,\ \ \vec{E}_s = -\frac{dV}{ds}, V(r) = \frac{1}{4\pi\epsilon_0}\frac{Q}{r}$$
$$\Phi = E\cdot dA = Ecos(\theta)dA = \oint \vec{E}Cos(\theta)dA = \frac{q_{enclosed}}{\epsilon_0}$$
$$U = \frac{q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}\epsilon V_{Vol}E^2,\ \ U_B = \frac{1}{2}Li^2,\ \ \omega = \frac{1}{\sqrt{LC}},$$
$$\sum I_{Junc} = 0,\ \ \sum V_{loop} = 0$$

$$R_T = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1}(P), C_T = \left(\frac{1}{C_1} + \frac{1}{C_2}\right)^{-1}(S)$$
$$\vec{F}_B = q\left(\vec{E} + \vec{v}\times\vec{B}\right) = I\vec{L}\times\vec{B} = \frac{mv^2}{r} = \frac{\mu_0 l I_1 I_2}{2\pi d}$$
$$X = Ae^{-\frac{t}{RC}} = A\left(1 - e^{-\frac{t}{RC}}\right), \tau = RC$$
$$B = \frac{\mu_0 I}{2\pi R} = \frac{\Phi_B}{ACos(\theta)}, \varepsilon = -N\frac{d\Phi_B}{dt}$$
$$\varepsilon = -L\frac{dI}{dt}, U_B = \frac{1}{2}Li^2, L = \frac{N\Phi_B}{I}$$
$$X_c = \frac{1}{\omega c}, Z_c = -\frac{j}{\omega c}, X_L = \omega L, Z_L = j\omega L,\ \ pf = Cos(\phi)$$
$$S = P + jQ, P_{Ave} = V_{Rms}I_{Rms}Cos(\phi) = \frac{1}{2}V_0 I_0 Cos(\phi)$$

$$\Delta E = E_{High} - E_{Low} = \frac{h^2}{8ml^2}(n_{High}^2 - n_{Low}^2)$$
$$r_n = \frac{\epsilon_0 h^2}{\pi m_e e^2}n^2 = an^2, a = 5.29 \times 10^{-11}[m]$$
$$\frac{1}{\lambda} = R\left(\frac{1}{n_{Low}^2} - \frac{1}{n_{High}^2}\right), R = 1.097 \times 10^7 [m^{-1}]$$
$$Orbitals: s, p, d, f, g \rightarrow \ell = 0,1,2,3,4,\cdots$$

$$\psi(r) = \frac{1}{\sqrt{\pi}}a^{-\frac{3}{2}}e^{-\frac{r}{a}}\ (H\ Gnd\ state)$$
$$\psi^2(r)dV = \frac{4}{a^3}e^{-\frac{2r}{a}}r^2 dr$$

$1s$,
$2s\ 2p$,
$3s\ 3p\ 3d$,
$4s\ 4p\ 4d\ 4f$,
$5s\ 5p\ 5d\ 5f$,
$6s\ 6p\ 6d$,
$7s\ 7p$,
$2, 6, 10, 14$

$$S = \sqrt{s(s+1)}\hbar = \frac{\sqrt{3}}{2}\hbar, \mu_s = -\frac{\ell}{m}S$$
$$1u = \frac{931.494013MeV}{c^2},\ \ \Delta = M - \sum m$$
$$E_{be} = \sum(mc^2) - Mc^2 = (\Delta m)c^2,\ \ \Delta E_{ben} = \frac{\Delta E_{be}}{A}$$
$$\alpha: {}^A_Z X \rightarrow {}^{A-4}_{Z-2}Y + {}^4_2 He + E$$
$$\gamma: {}^A_Z X^* \rightarrow {}^A_Z X + \gamma + E$$
$$N = N_0 e^{-\lambda t}, t_{\frac{1}{2}} = \frac{\ln(2)}{\lambda}$$
$$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 Nm^2 C^{-2}$$
$$I = \frac{dq}{dt} = \frac{q}{t},\ \ \vec{E} = \frac{\vec{F}}{q_0} = \frac{Q}{4\pi\epsilon_0 r^2}\hat{r}$$
$$\nabla\times\vec{E} = 0,\ \ W = \int_1^2 F\cdot dx, U = -W = -\int_\infty^2 \vec{F}\cdot ds$$
$$\vec{\tau} = \vec{p}\times\vec{E}, U = -\vec{p}\cdot\vec{E}, E = \frac{\sigma}{2\epsilon_0}\left(1 - \frac{z}{\sqrt{z^2+r^2}}\right)$$
$$E = \frac{\sigma}{2\epsilon_0}(\infty\ Sheet),\ \ C = \frac{Q}{V} = \frac{\epsilon_0\kappa A}{d},\ \ \vec{E} = \frac{V}{d}$$
$$P = VI = I^2R = \frac{V^2}{R},\ \ V = IR,\ \ R = \frac{\rho l}{A}$$
$$V_1 = V\frac{R_1}{R_1+R_2}, I_1 = I_t\frac{R_2}{R_1+R_2}$$
$$\varepsilon = V - IR,\ \ R_0 = \frac{\varepsilon}{I_{sc}} = R_L\left(\frac{\varepsilon}{V} - 1\right)$$
$$\tau = IABSin(\theta) = I\vec{A}\times\vec{B}$$

$$\Phi_B = \oint \vec{B}\cdot d\vec{A} = BA$$
$$\oint E\cdot dA = \frac{q_{enc}}{\epsilon_0},\ \ \oint B\cdot dA = 0$$
$$\oint E\cdot ds = -\frac{d\Phi_B}{dt},\ \ \oint B\cdot ds = \mu_0\epsilon_0\frac{d\Phi_E}{dt} + \mu_0 I_{enc}$$